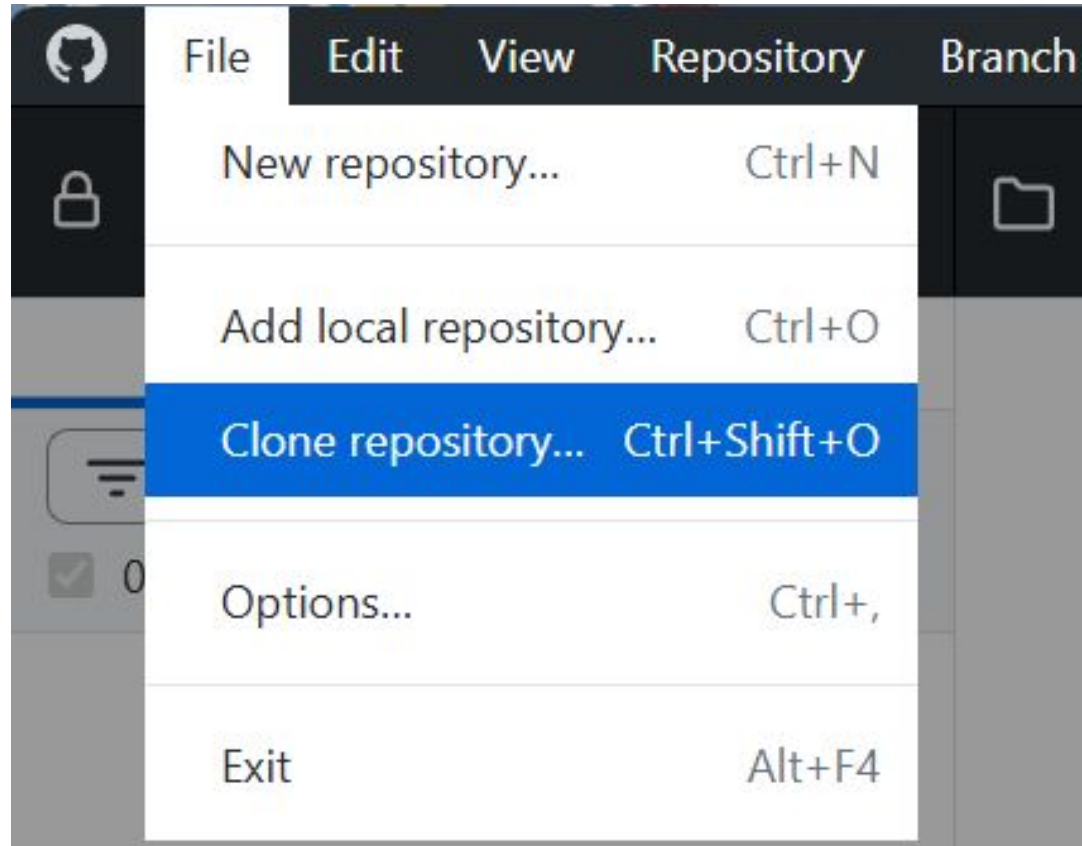


Github desktop + VS Code setup

<https://desktop.github.com/download/> - GitHub Desktop



Clone via public repo URL:

<https://github.com/YurySergeev/greenhouse-monitoring-system-year1>

Choose a path
you know works →

Clone a repository

GitHub.com

GitHub Enterprise

URL

Repository URL or GitHub username and repository
(hubot/cool-repo)

https://github.com/YurySergeev/greenhouse-monitoring-system-year1

Local path

C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-y

Choose...

Clone

Cancel



Current branch
main



Fetch origin
Never fetched

Branches

Pull requests



Filter

New branch

Default branch



main

6 months ago

Other branches



origin/37-fe---implement-auto-refresh

7 months ago



origin/Dev

5 months ago



origin/alert-system

5 months ago



origin/api-connection

7 months ago



origin/cassini/work

5 months ago

Set Dev branch





Current branch
Dev



Fetch origin
Never fetched

No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.



Create a Pull Request from your current branch

The current branch (Dev) is already published to GitHub. Create a pull request to propose and collaborate on your changes.

Branch menu or `Ctrl + R`

Create Pull Request

Open the repository in your external editor

Select your editor in [Options](#)

Repository menu or `Ctrl + Shift + A`

Open in Visual Studio Code

View the files of your repository in Explorer

Repository menu or `Ctrl + Shift + F`

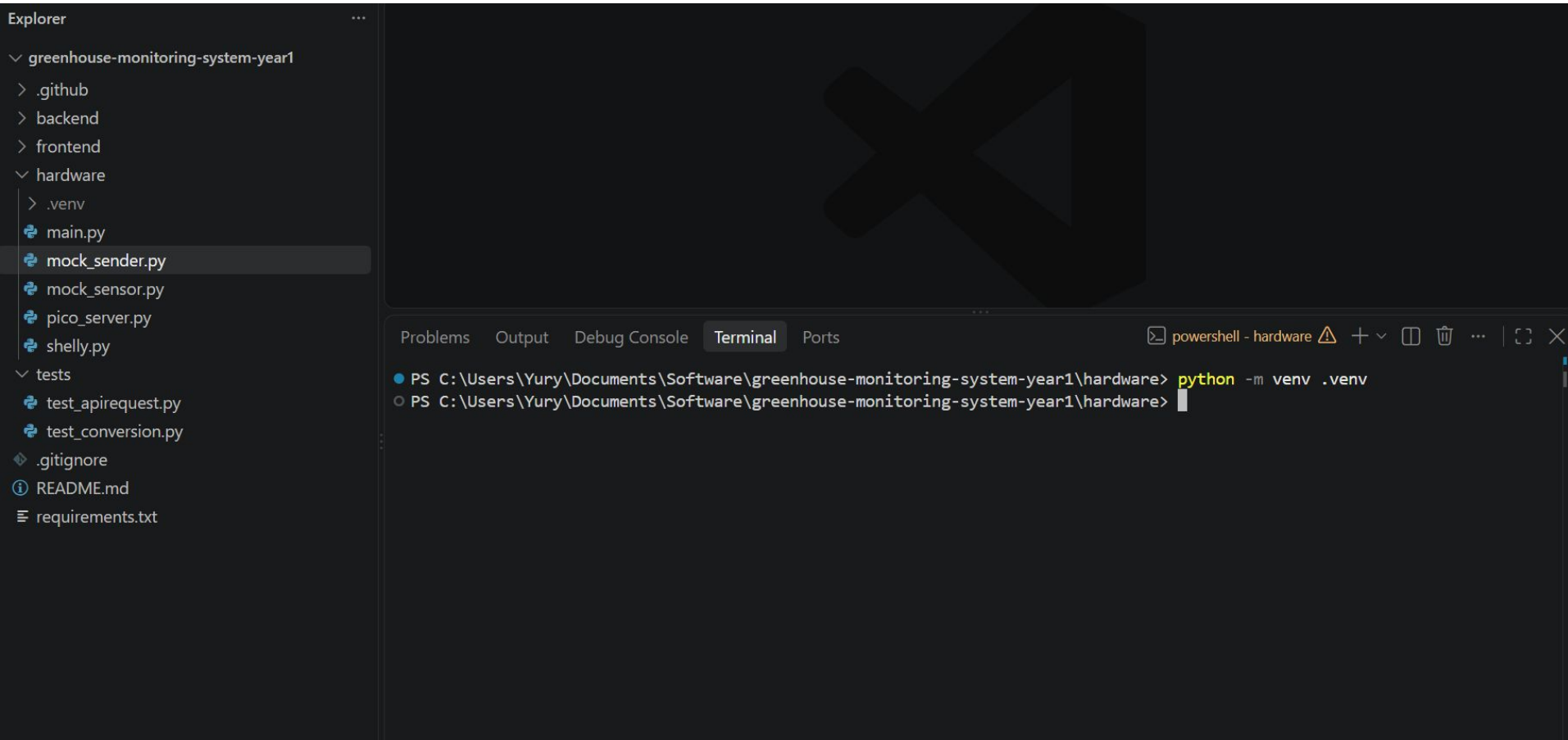
Show in Explorer

Open the repository page on GitHub in your browser

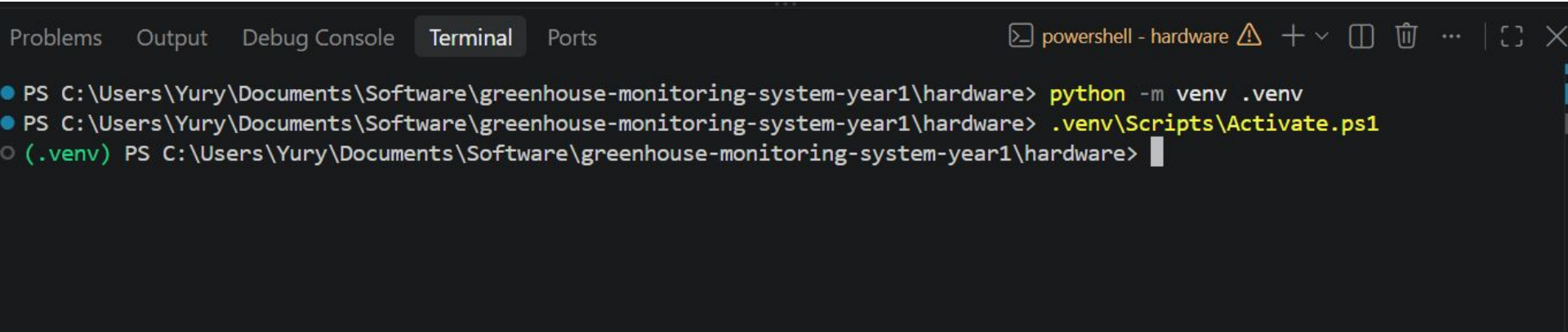
Repository menu or `Ctrl + Shift + G`

View on GitHub

cd hardware first then run .venv



Activate script



The screenshot shows a Visual Studio Code interface with a terminal window open. The terminal title bar indicates it is a PowerShell session in the 'hardware' directory. The terminal content shows three commands being executed in sequence: first, 'python -m venv .venv' to create the virtual environment; second, '.venv\Scripts\Activate.ps1' to activate it; and third, the prompt changes to '(.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1\hardware>', indicating the virtual environment is now active. The terminal window has a dark background with light-colored text. The Visual Studio Code interface includes tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports' at the top. The terminal window also has standard icons for window management (maximize, close, etc.) on the right side of the title bar.

```
Problems Output Debug Console Terminal Ports powershell - hardware ⚠ + ▾ 🗑 ... | 📄 ✕  
● PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1\hardware> python -m venv .venv  
● PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1\hardware> .venv\Scripts\Activate.ps1  
○ (.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1\hardware> |
```

cd .. (back into main folder)

```
● (.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1\hardware> cd ..
○ (.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1> pip install -r requirements.txt
Collecting flask (from -r requirements.txt (line 2))
  Using cached flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting streamlit (from -r requirements.txt (line 3))
  Downloading streamlit-1.64.0-py3-none-any.whl.metadata (10 kB)
Collecting streamlit-autorefresh (from -r requirements.txt (line 4))
  Using cached streamlit_autorefresh-1.0.1-py3-none-any.whl.metadata (436 bytes)
Collecting matplotlib (from -r requirements.txt (line 5))
  Downloading matplotlib-3.11.2-cp311-cp311-win_amd64.whl.metadata (80 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 80.3/80.3 kB 1.1 MB/s eta 0:00:00
```

This will take a while - some libs as big as ~200mb

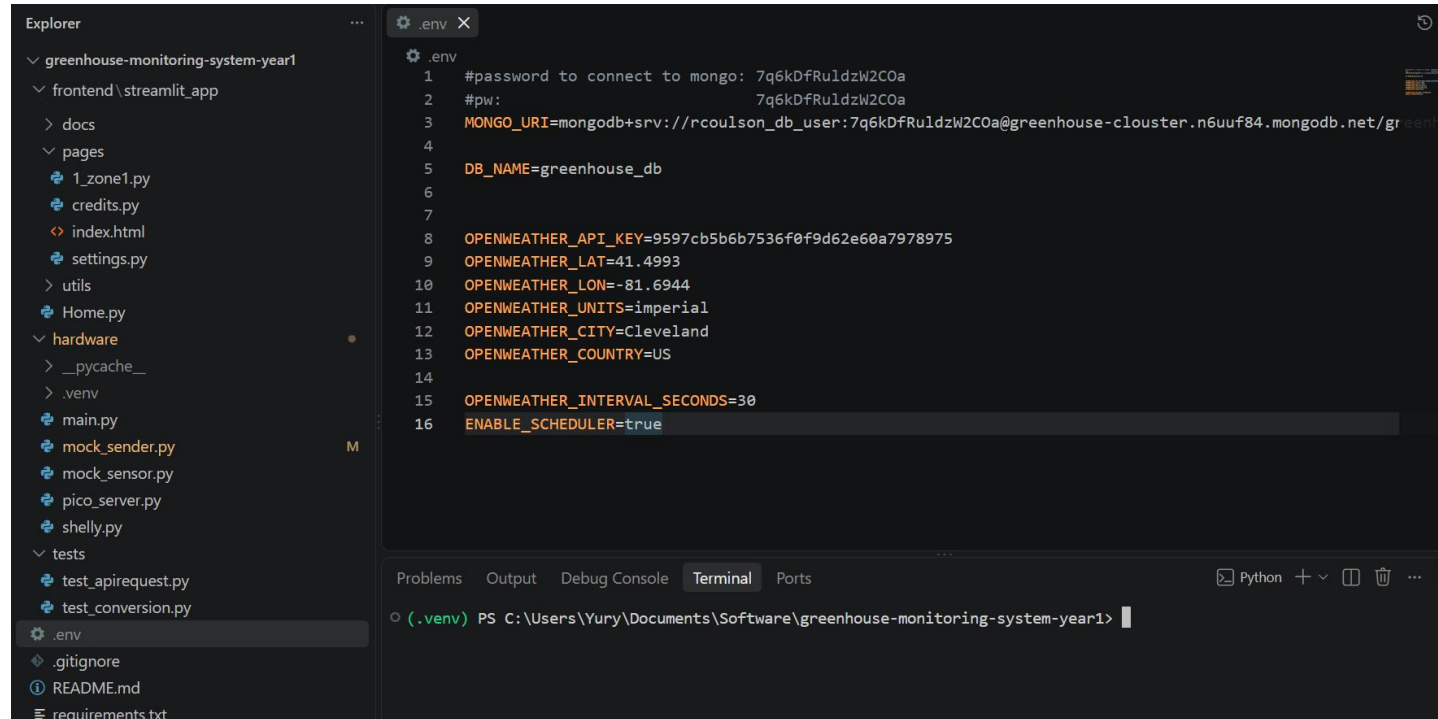
- **Wait until done, restart VS**

Create a .env file in
main folder

>click create new
file

>name it .env

>copy provided .env
copy



The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays the file structure of a project named 'greenhouse-monitoring-system-year1'. The file '.env' is highlighted in the list. The main editor window shows the content of the '.env' file, which contains configuration for a MongoDB database and OpenWeather API. The terminal at the bottom shows the command prompt for the '.venv' environment.

```
.env
1 #password to connect to mongo: 7q6kDfRu1dzW2C0a
2 #pw: 7q6kDfRu1dzW2C0a
3 MONGO_URI=mongodb+srv://rcoulson_db_user:7q6kDfRu1dzW2C0a@greenhouse-clouster.n6uuf84.mongodb.net/greenhouse-monitoring-system-year1
4
5 DB_NAME=greenhouse_db
6
7
8 OPENWEATHER_API_KEY=9597cb5b6b7536f0f9d62e60a7978975
9 OPENWEATHER_LAT=41.4993
10 OPENWEATHER_LON=-81.6944
11 OPENWEATHER_UNITS=imperial
12 OPENWEATHER_CITY=Cleveland
13 OPENWEATHER_COUNTRY=US
14
15 OPENWEATHER_INTERVAL_SECONDS=30
16 ENABLE_SCHEDULER=true
```

Terminal:

```
(.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1>
```

>Open Windows cmd

>ipconfig

Find IPv4 Address

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Yury>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : attlocal.net
    IPv6 Address. . . . . : 2600:1700:94c0:12d0::22
    IPv6 Address. . . . . : 2600:1700:94c0:12d0:9169:80d1:b74d:3db4
    Temporary IPv6 Address. . . . . : 2600:1700:94c0:12d0:8d91:e1de:278e:b8c1
    Link-local IPv6 Address . . . . . : fe80::3d54:944a:a2c7:a7df%8
    IPv4 Address. . . . . : 192.168.1.189
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::9a40:d4ff:fe0a:fb44%8
                               192.168.1.254
```

mock_sender.py M X

hardware > mock_sender.py > ...

```
22
23 SERVER_URL = "http://192.168.1.189:5000/data"
24 INTERVAL   = 5    # seconds between each reading
25
26
```

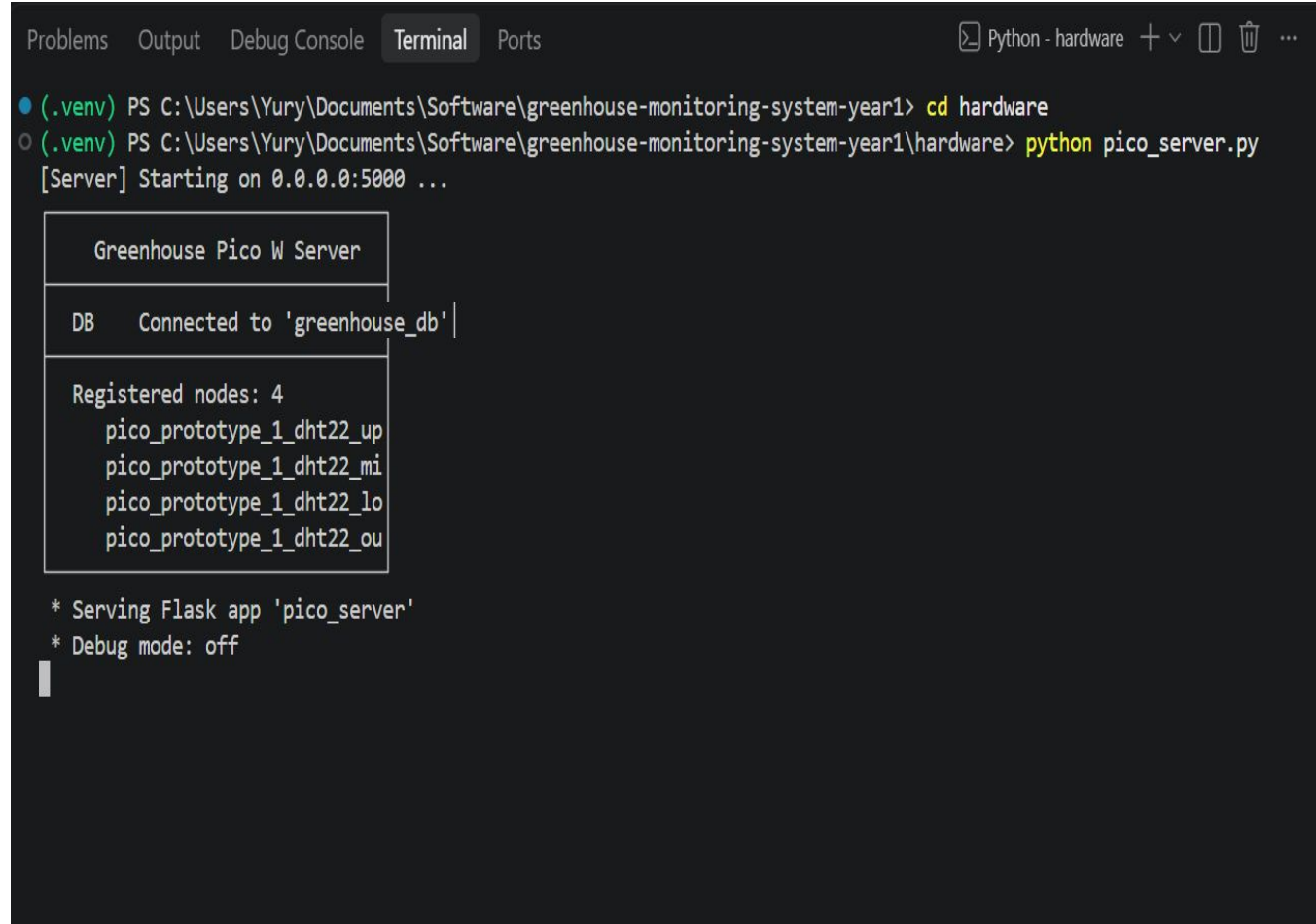
Replace SERVER_URL ip address in hardware/mock_sender.py

To run the server on
your machine

>cd hardware

>python pico_server.py

Connects to DB only if
you have the .env setup



```
Problems Output Debug Console Terminal Ports Python - hardware + v [] [X] ...
(.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1> cd hardware
(.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1\hardware> python pico_server.py
[Server] Starting on 0.0.0.0:5000 ...

Greenhouse Pico W Server

DB Connected to 'greenhouse_db'

Registered nodes: 4
  pico_prototype_1_dht22_up
  pico_prototype_1_dht22_mi
  pico_prototype_1_dht22_lo
  pico_prototype_1_dht22_ou

* Serving Flask app 'pico_server'
* Debug mode: off
```

```
• (.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1> cd hardware
• (.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1\hardware> python mock_sender.py
[Dummy Sender] Starting - POSTing to http://192.168.1.189:5000/data every 5s
[Dummy Sender] Press Ctrl+C to stop.

[Reading] {'node_id': 'pico_prototype_1_dht22_upper', 'zone': 'zone1', 'area': 'upper_plants', 'temperature_c': 22.15, 'humidity_rh': 60.64, 'status': 'ok_simulated'}
[Server] Saved - id: 6aa9c2b5c563dab7508aba71
[Waiting] Next reading in 5s...
```

Simulate sending of authorized node - if node name is bad it is declined

>cd hardware

>python mock_sender.py

* Second terminal

* will send random generated data

Server console view

Displays device, ip, zone, reading and where it is saved to

Saves based on name decode

```
[Server] Starting on 0.0.0.0:5000 ...

Greenhouse Pico W Server

DB    Connected to 'greenhouse_db'|

Registered nodes: 4
  pico_prototype_1_dht22_up
  pico_prototype_1_dht22_mi
  pico_prototype_1_dht22_lo
  pico_prototype_1_dht22_ou

* Serving Flask app 'pico_server'
* Debug mode: off
[18:12:05] ! New device connected
          ID      : pico_prototype_1_dht22_upper
          IP      : 192.168.1.189
          Zone/Area : zone1 / upper_plants
          Collection : zone1-upper

[18:12:05] pico_prototype_1_dht22_upper | 22.1°C | 61% RH | 192.168.1.189
[18:12:05] pico_prototype_1_dht22_upper | 22.1°C | 61% RH | #1
[18:12:05] Saved to 'zone1-upper'       | _id: 6aa9c2b5c563dab7508aba71
```

```
(.venv) PS C:\Users\Yury\Documents\Software\greenhouse-monitoring-system-year1\frontend\streamlit_app>
streamlit run Home.py
2026-09-15 18:13:37.901 Uvicorn server started on :::8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.189:8501

Help agents write better Streamlit apps?
Install the official Streamlit skills by running streamlit skills in your terminal.
```

To run frontend

```
> cd frontend/streamlit_app
```

```
> streamlit run Home.py
```

*First start can take a bit to render

Home

Climate

Settings



Greenhouse Console

Zone 1 climate is within target band

Upper canopy is reading 22.1°C and 61% RH. Outside reference is 20.7°C.

NAVIGATE

Home

Greenhouse Home

● Operations

A single place to watch canopy conditions, compare inside and outside climate, and spot issues before they stress the plants.

KENT STATE GREENHOUSE

Clean, fast climate awareness for daily greenhouse work.

This home screen keeps the most important context above the fold: live indoor climate, outside reference conditions, reporting sensor positions, and whether any area has drifted outside the configured plant-safe range.

4/4 positions reporting

0 canopy bands in target

No active climate exceptions



● Kent State greenhouse



Current Snapshot

ZONE 1 TEMPERATURE

22.1°C

Upper canopy live reading.

ZONE 1 HUMIDITY

61% RH

Relative humidity at the primary canopy monitor.

OUTSIDE REFERENCE

20.7°C

Weather feed used as a greenhouse reference point.

ATTENTION

Clear

All monitored indoor positions are inside the target band.



Monitoring Positions

POSITION

● Outside reference

Upper canopy

22.1°C and 61% RH

POSITION

● Outside reference

Mid canopy

22.5°C and 60% RH

POSITION

● Outside reference

Root zone

21.4°C and 60% RH

NAVIGATE

Home

Zone 1 Dashboard

Zone 2

Credits

CONTROLS

Alert Settings

Glasshouse 

Live snapshot from Zone 1 upper canopy

Temperature, last 24 hours



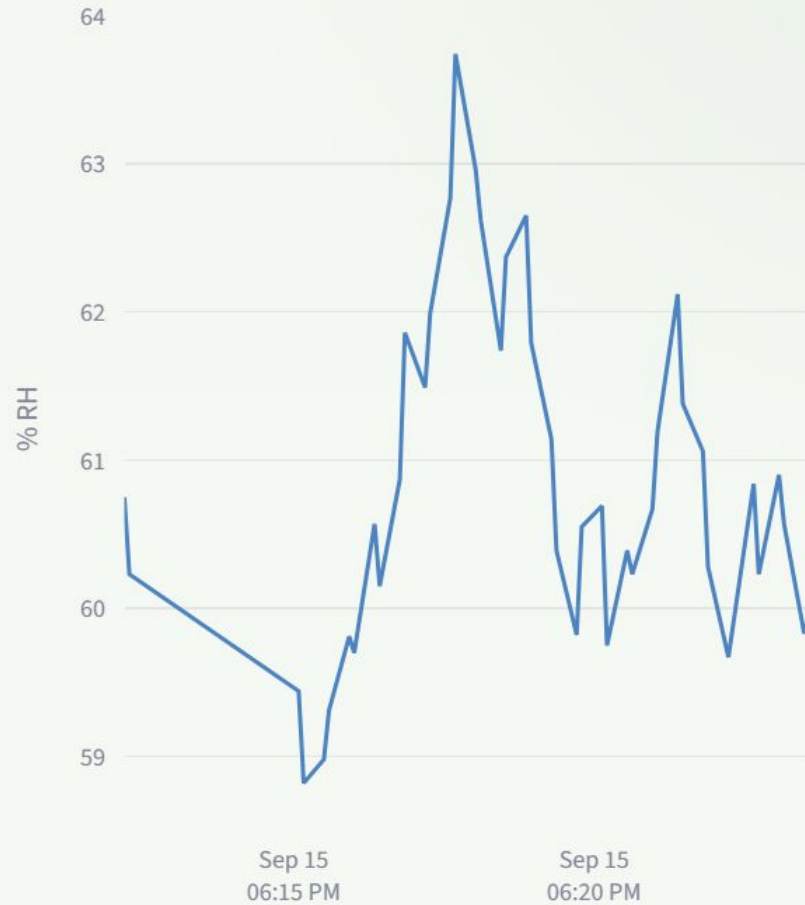
Humidity, last 24 hours



Temperature, last 24 hours



Humidity, last 24 hours



Temperature, last 24 hours



Humidity, last 24 hours

